

CS 441 Discrete Mathematics for CS Lecture 3

Predicate logic

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Propositional logic: review

- **Propositional logic:** a formal language for making logical inferences
- A **proposition** is a statement that is either true or false.
- A **compound proposition** can be created from other propositions using logical connectives
- **The truth of a compound proposition** is defined by truth values of elementary propositions and the meaning of connectives.
- **The truth table for a compound proposition:** table with entries (rows) for all possible combinations of truth values of elementary propositions.

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Tautology and Contradiction

What is a tautology?

- A compound proposition that is always true for all possible truth values of the propositions is called a **tautology**.
-

Example: $p \vee \neg p$ is a **tautology**.

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Tautology and Contradiction

What is a contradiction?

- A compound proposition that is always false is called a **contradiction**.

Example: $p \wedge \neg p$ is a **contradiction**.

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Equivalence

- How do we determine that two propositions are equivalent?
Their truth values in the truth table are the same.
- Example: $p \rightarrow q$ is equivalent to $\neg q \rightarrow \neg p$ (**contrapositive**)

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

- Equivalent statements** are important for logical reasoning since they can be substituted and can help us to make a logical argument.

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Logical equivalence

- Definition:** The propositions p and q are called **logically equivalent** if $p \leftrightarrow q$ is a tautology (alternately, if they have the same truth table). The notation $p \Leftrightarrow q$ denotes p and q are logically equivalent.

a	b	$a \rightarrow b$	$\neg a \rightarrow \neg b$	$(a \rightarrow b) \leftrightarrow (\neg a \rightarrow \neg b)$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

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Important logical equivalences

- **Identity**

- $p \wedge T \Leftrightarrow p$
- $p \vee F \Leftrightarrow p$

- **Domination**

- $p \vee T \Leftrightarrow T$
- $p \wedge F \Leftrightarrow F$

- **Idempotent**

- $p \vee p \Leftrightarrow p$
- $p \wedge p \Leftrightarrow p$

Important logical equivalences

- **Double negation**

- $\neg(\neg p) \Leftrightarrow p$

- **Commutative**

- $p \vee q \Leftrightarrow q \vee p$
- $p \wedge q \Leftrightarrow q \wedge p$

- **Associative**

- $(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$
- $(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$

Important logical equivalences

- **Distributive**

- $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
- $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

- **De Morgan**

- $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

- **Other useful equivalences**

- $p \vee \neg p \Leftrightarrow T$
- $p \wedge \neg p \Leftrightarrow F$
- $p \rightarrow q \Leftrightarrow (\neg p \vee q)$

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Showing logical equivalence

Example: Show $(p \wedge q) \rightarrow p$ is a tautology

In other words $((p \wedge q) \rightarrow p \Leftrightarrow T)$

Proof via truth table:

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

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Showing logical equivalences

- **Equivalences can be used in proofs.** A proposition or its part can be transformed using equivalences and some conclusion can be reached.

Example: Show that $(p \wedge q) \rightarrow p$ is a tautology.

- Proof: (we must show $(p \wedge q) \rightarrow p \Leftrightarrow T$)
$$\begin{aligned}(p \wedge q) \rightarrow p &\Leftrightarrow \neg(p \wedge q) \vee p && \text{Useful} \\ &\Leftrightarrow [\neg p \vee \neg q] \vee p && \text{DeMorgan} \\ &\Leftrightarrow [\neg q \vee \neg p] \vee p && \text{Commutative} \\ &\Leftrightarrow \neg q \vee [\neg p \vee p] && \text{Associative} \\ &\Leftrightarrow \neg q \vee [T] && \text{Useful} \\ &\Leftrightarrow T && \text{Domination}\end{aligned}$$

Propositional logic

- **Definition:**
 - A **proposition** is a statement that is either true or false.
- **Examples:**
 - Pitt is located in the Oakland section of Pittsburgh.
 - $5 + 2 = 8$.
 - It is raining today
 - 2 is a prime number
 - If (you do not drive over 65 mph) then (you will not get a speeding ticket).
- **Not a proposition:**
 - How are you?
 - $x + 5 = 3$

Limitations of the propositional logic

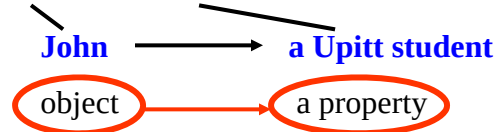
Propositional logic: the world is described in terms of elementary propositions and their logical combinations

Elementary statements:

- Typically refer to objects, their properties and relations.
But these are not explicitly represented in the propositional logic

– **Example:**

- “John is a UPitt student.”



- Objects and properties are hidden in the statement, it is not possible to reason about them

Limitations of the propositional logic

(1) Statements that must be repeated for many objects

- In propositional logic these must be exhaustively enumerated

• **Example:**

- If John is a CS UPitt graduate then John has passed cs441

Translation:

- John is a CS UPitt graduate $\rightarrow$ John has passed cs441

Similar statements can be written for other Upitt graduates:

- Ann is a CS Upitt graduate $\rightarrow$ Ann has passed cs441

- Ken is a CS Upitt graduate $\rightarrow$ Ken has passed cs441

– ...

- What is a more natural solution to express the above knowledge?

Limitations of the propositional logic

(1) Statements that must be repeated for many objects

- **Example:**

- If John is a CS UPitt graduate then John has passed cs441

Translation:

- John is a CS UPitt graduate $\rightarrow$ John has passed cs441

Similar statements can be written for other Upitt graduates:

- Ann is a CS Upitt graduate $\rightarrow$ Ann has passed cs441
- Ken is a CS Upitt graduate $\rightarrow$ Ken has passed cs441
- ...

- **Solution:** make statements with **variables**

- If x is a CS Upitt graduate then x has passed cs441
- x is a CS UPitt graduate $\rightarrow x$ has passed cs441

Limitations of the propositional logic

(2) Statements that define the property of the group of objects

- **Example:**

- All new cars must be registered.
- Some of the CS graduates graduate with honors.

- **Solution:** make statements with **quantifiers**

- **Universal quantifier** – the property is satisfied by all members of the group
- **Existential quantifier** – at least one member of the group satisfy the property

Predicate logic

Remedies the limitations of the propositional logic

- Explicitly models objects and their properties
- Allows to make statements with variables and quantify them

Basic building blocks of the predicate logic:

- **Constant** –models a specific object
Examples: “John”, “France”, “7”
- **Variable** – represents object of specific type (**defined by the universe of discourse**)
Examples: x, y
(universe of discourse can be people, students, numbers)
- **Predicate** - over one, two or many variables or constants.
 - Represents properties or relations among objects**Examples:** Red(car23), student(x), married(John,Ann)

Predicates

Predicates represent properties or relations among objects

A predicate $P(x)$ assigns a value **true or false** to each x depending on whether the property holds or not for x .

- The assignment is best viewed as a big table with the variable x substituted for objects from *the universe of discourse*

Example:

- Assume **Student(x)** where the universe of discourse are people
- Student(John) T (if John is a student)
- Student(Ann) T (if Ann is a student)
- Student(Jane) F (if Jane is not a student)
- ...

Predicates

Assume a predicate $P(x)$ that represents the statement:

- x is a prime number

What are the truth values of:

- | | |
|----------|---|
| • $P(2)$ | T |
| • $P(3)$ | T |
| • $P(4)$ | F |
| • $P(5)$ | T |
| • $P(6)$ | F |
| • $P(7)$ | T |

All statements $P(2)$, $P(3)$, $P(4)$, $P(5)$, $P(6)$, $P(7)$ are propositions

Predicates

Assume a predicate $P(x)$ that represents the statement:

- x is a prime number

What are the truth values of:

- | | |
|----------|---|
| • $P(2)$ | T |
| • $P(3)$ | T |
| • $P(4)$ | F |
| • $P(5)$ | T |
| • $P(6)$ | F |
| • $P(7)$ | T |

Is $P(x)$ a proposition? No. Many possible substitutions are possible.

Predicates

- Predicates can have **more arguments** which represent the **relations between objects**

Example:

- $\text{Older}(\text{John}, \text{Peter})$ denotes 'John is older than Peter'
 - this is a proposition because it is either true or false
- $\text{Older}(x, y)$ - 'x is older than y'
 - not a proposition, but after the substitution it becomes one

Predicates

- Predicates can have **more arguments** which represent the **relations between objects**

Example:

- Let $Q(x, y)$ denote ' $x+5 > y$ '
 - Is $Q(x, y)$ a proposition?

Predicates

- Predicates can have **more arguments** which represent the **relations between objects**

Example:

- Let $Q(x,y)$ denote ' $x+5 > y$ '
 - Is $Q(x,y)$ a proposition? **No!**
 - Is $Q(3,7)$ a proposition? **Yes.** It is true.
 - What is the truth value of:
 - $Q(3,7)$ T
 - $Q(1,6)$ F
 - $Q(2,2)$ T
 - Is $Q(3,y)$ a proposition? **No!** We cannot say if it is true or false.

Compound statements in predicate logic

Compound statements are obtained via logical connectives

Examples:

$\text{Student}(\text{Ann}) \wedge \text{Student}(\text{Jane})$

- **Translation:** "Both Ann and Jane are students"
- **Proposition:** yes.

$\text{Country}(\text{Sienna}) \vee \text{River}(\text{Sienna})$

- **Translation:** "Sienna is a country or a river"
- **Proposition:** yes.

$\text{CS-major}(x) \rightarrow \text{Student}(x)$

- **Translation:** "if x is a CS-major then x is a student"
- **Proposition:** no.

Predicates

Important:

- statement $P(x)$ is **not a proposition** since there are more objects it can be applied to

This is the same as in propositional logic ...

... But the difference is:

- predicate logic allows us to explicitly manipulate and substitute for the objects
- Predicate logic permits quantified sentences where variables are substituted for statements about the group of objects

Quantified statements

Predicate logic lets us to make statements about groups of objects

- To do this we use special quantified expressions

Two types of quantified statements:

- **universal**

Example: ‘all CS Upitt graduates have to pass cs441’

– the statement is true for all graduates

- **existential**

Example: ‘Some CS Upitt students graduate with honor.’

– the statement is true for some people

Universal quantifier

Defn: The universal quantification of $P(x)$ is the proposition:

" $P(x)$ is true for all values of x in the domain of discourse." The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$, and is expressed as **for every x , $P(x)$** .

Example:

- Let $P(x)$ denote $x > x - 1$.
- What is the truth value of $\forall x P(x)$?
- Assume the universe of discourse of x is all real numbers.
- **Answer:** Since every number x is greater than itself minus 1. Therefore, **$\forall x P(x)$ is true.**

Universal quantifier

Quantification converts a propositional function into a **proposition** by binding a variable to a set of values from the universe of discourse.

Example:

- Let $P(x)$ denote $x > x - 1$.
- Is $P(x)$ a proposition? **No.** Many possible substitutions.
- Is $\forall x P(x)$ a proposition? **Yes.** True if for all x from the universe of discourse $P(x)$ is true.

Universally quantified statements

Predicate logic lets us make statements about groups of objects

Universally quantified statement

- $\text{CS-major}(x) \rightarrow \text{Student}(x)$
 - **Translation:** “if x is a CS-major then x is a student”
 - **Proposition:** **no.**
- $\forall x \text{CS-major}(x) \rightarrow \text{Student}(x)$
 - **Translation:** “(For all people it holds that) if a person is a CS-major then she is a student.”
 - **Proposition:** **yes.**

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Existential quantifier

Definition: The **existential quantification** of $P(x)$ is the proposition “*There exists an element in the domain (universe) of discourse such that $P(x)$ is true.*” The notation $\exists x P(x)$ denotes the existential quantification of $P(x)$, and is expressed as **there is an x such that $P(x)$ is true.**

Example 1:

- Let $T(x)$ denote $x > 5$ and x is from Real numbers.
- What is the truth value of $\exists x T(x)$?
- **Answer:**
- Since $10 > 5$ is true. Therefore, it is **true that $\exists x T(x)$.**

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Existential quantifier

Definition: The **existential quantification** of $P(x)$ is the proposition "*There exists an element in the domain (universe) of discourse such that $P(x)$ is true.*" The notation $\exists x P(x)$ denotes the existential quantification of $P(x)$, and is expressed as **there is an x such that $P(x)$ is true.**

Example 2:

- Let $Q(x)$ denote $x = x + 2$ where x is real numbers
- What is the truth value of $\exists x Q(x)$?
- **Answer:** Since no real number is 2 larger than itself, the truth value of $\exists x Q(x)$ is false.

Existentially quantified statements

Statements about groups of objects

Example:

- $\text{CS-UPitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “ x is a CS-UPitt-graduate and x is an honor student”
 - **Proposition:** ?

Quantified statements

Statements about groups of objects

Example:

- $\text{CS-Upitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “x is a CS-Upitt-graduate and x is an honor student”
 - **Proposition:** **no.**
- $\exists x \text{CS-Upitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “There is a person who is a CS-Upitt-graduate and who is also an honor student.”
 - **Proposition:** ?

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Quantified statements

Statements about groups of objects

Example:

- $\text{CS-Upitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “x is a CS-Upitt-graduate and x is an honor student”
 - **Proposition:** **no.**
- $\exists x \text{CS-Upitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “There is a person who is a CS-Upitt-graduate and who is also an honor student.”
 - **Proposition:** **yes.**

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Summary of quantified statements

- When $\forall x P(x)$ and $\exists x P(x)$ are true and false?

Statement	When true?	When false?
$\forall x P(x)$	$P(x)$ true for all x	There is an x where $P(x)$ is false.
$\exists x P(x)$	There is some x for which $P(x)$ is true.	$P(x)$ is false for all x .

Suppose the elements in the universe of discourse can be enumerated as $x_1, x_2, \dots, x_N$ then:

- $\forall x P(x)$ is true whenever $P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_N)$ is true
- $\exists x P(x)$ is true whenever $P(x_1) \vee P(x_2) \vee \dots \vee P(x_N)$ is true.